STACKPILOT AI ENTERPRISE PLATFORM

Unified Operations & Business Intelligence OS

Bridging Technical Engineering, CRM, Automated Billing, and AI-Driven Execution in One Cohesive Architecture.

7 Workflow States
From Backlog to QA & Delivery

300ms Search
Debounced Instant Auto-complete

Bimodal Resilience
Express/Mongo + Offline Continuity

EXECUTIVE BRIEFING

The Enterprise Operational Challenge & Solution

THE SPREAD & SILO PROBLEM

• Engineering Disconnect: Developers operate in technical jargon without clear business requirement mappings.

• Opaque Delivery Status: Client sponsors lack real-time visibility into active sprints and velocity.

• Manual Billing Friction: Finance teams waste hours calculating milestone hours, tax, and PDF invoices.

• QA Specification Gaps: Testers lack structured acceptance criteria and automated test case scaffolding.

THE STACKPILOT AI SOLUTION

• Unified Workspaces: Plain-language business boards connect executives, developers, and clients seamlessly.

• Real-Time Telemetry: Transparent milestone metrics, automated sprint reports, and SLA health tracking.

• Instant Financial Billing: Automated invoice generator with customizable 18% GST/Tax and instant export.

• AI Generative Acceleration: Integrated AI tool suite for automated test cases, bug reports, and estimates.

TEAM GOVERNANCE

Specialized Personas & Role-Based Access (RBAC)

👑 Super Administrator
Security Audit & System Health — Complete access to telemetry, global user management, audit logs, and infrastructure configurations.

👔 Project Delivery Lead
Sprints, Scope & Approvals — Monitors task pipelines, sprint velocity, milestone deadlines, and scope change requests.

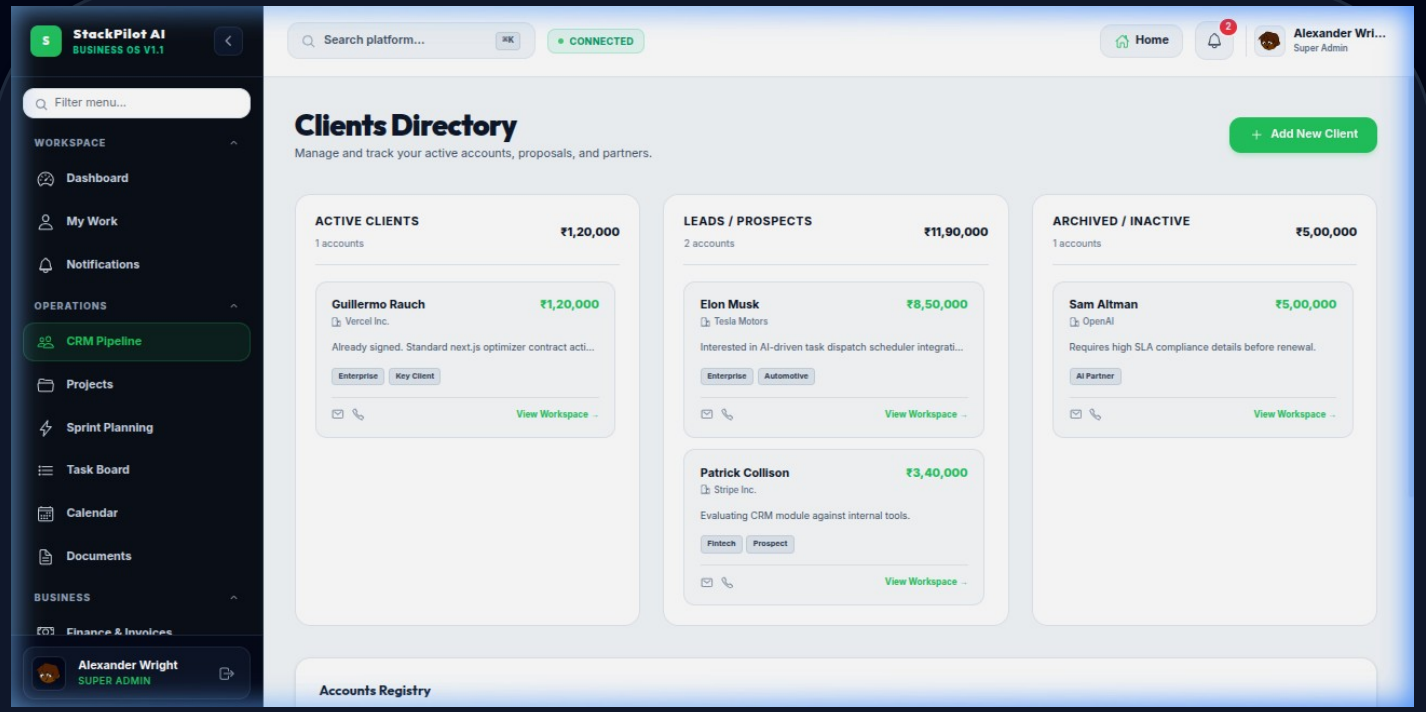
💻 Lead Developer
7-Column Board & AI Tools — Dedicated agile boards, automated code review assistants, and interactive technical documentation.

🧪 QA Test Engineer
UAT Scenarios & Bug Audits — QA column management, automated test case generator, and verification telemetry logs.

📊 Finance Coordinator
Invoices, Tax & Client SLAs — Automated invoice dispatch, 18% tax calculation, client retainer tracking, and balance reporting.

CUSTOMER RELATIONSHIP MANAGEMENT

CRM & Client Lifecycle Hub

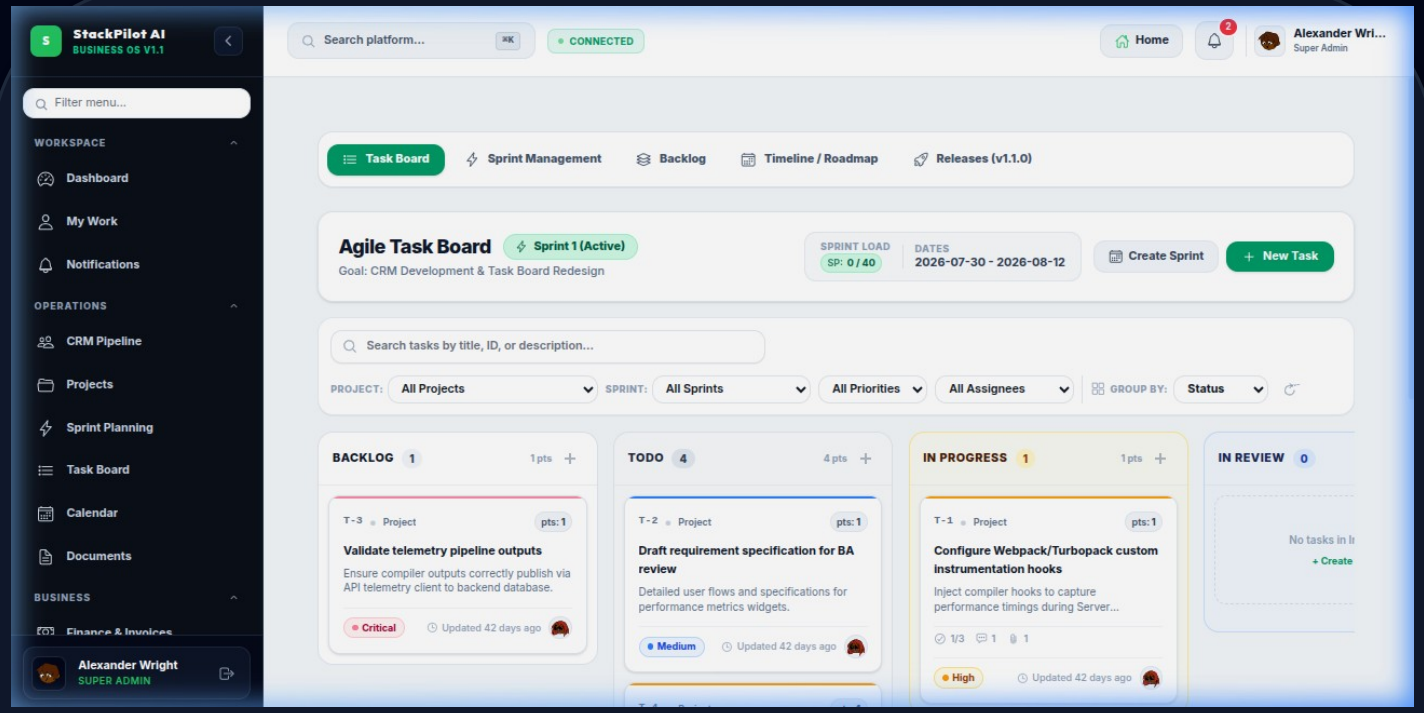


KEY CAPABILITIES

- **Active Accounts Registry**
  Real-time visibility into client retainers, active agreements, and lifetime value metrics.
- **Leads & Prospects Board**
  Direct management of high-value deals with instant status tracking and conversion.
- **1-Click Project Conversion**
  Instantly spin up dedicated workspace projects from closed client deals.
- **SLA & Contract Alerts**
  Automated flags for pending contract renewals and client SLA deliverables.

ENGINEERING EXECUTION

7-Column Agile Task Board & Sprint Engine

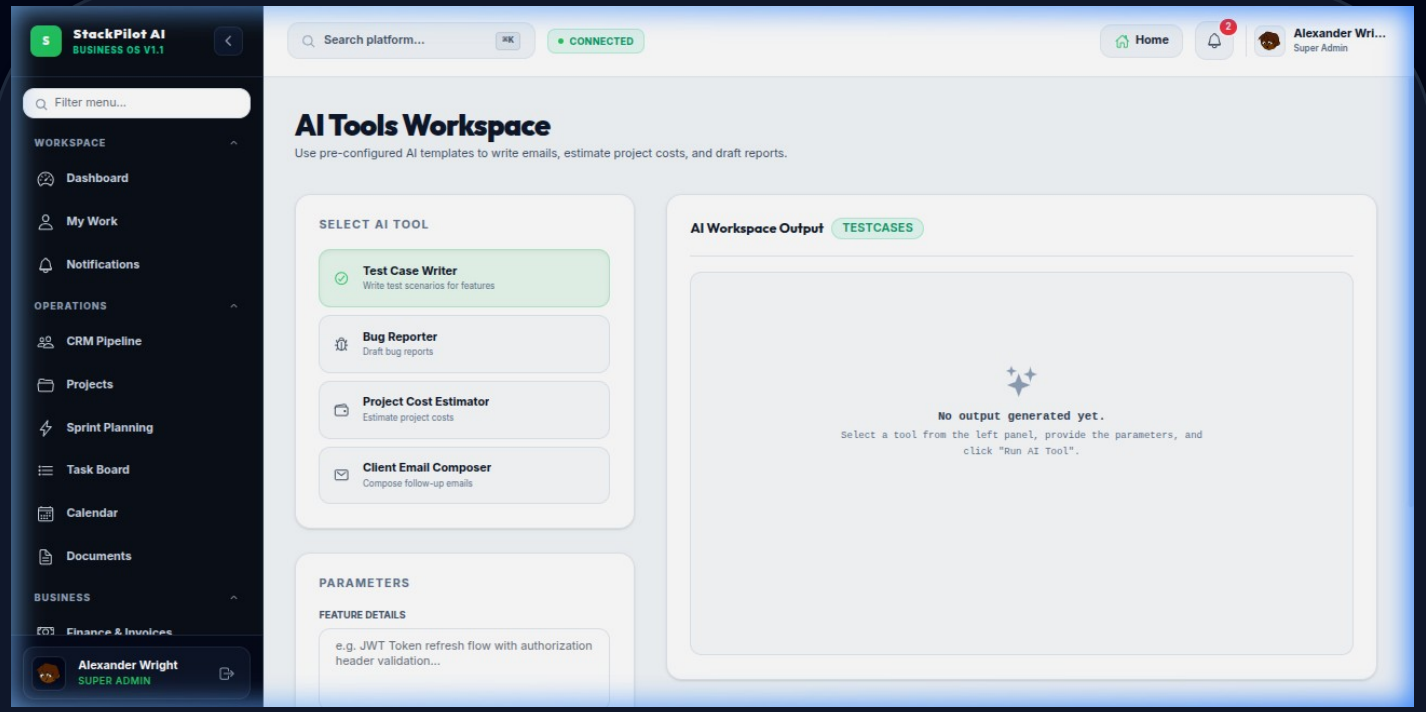


KEY CAPABILITIES

- **7 Granular Workflow Columns**
  Backlog, Todo, In Progress, In Review, QA Testing, and Completed states.
- **Sprint Capacity Telemetry**
  Real-time story point tracking and workload balancing per team member.
- **Multi-Dimensional Toolbar**
  Instant filtering across Projects, Sprints, Priority, Assignees, or Group-By.
- **Drag-and-Drop Experience**
  Fluid 60fps micro-animated task status transitions with audit tracking.

GENERATIVE AUTOMATION

AI Productivity Suite & Engineering Assistants

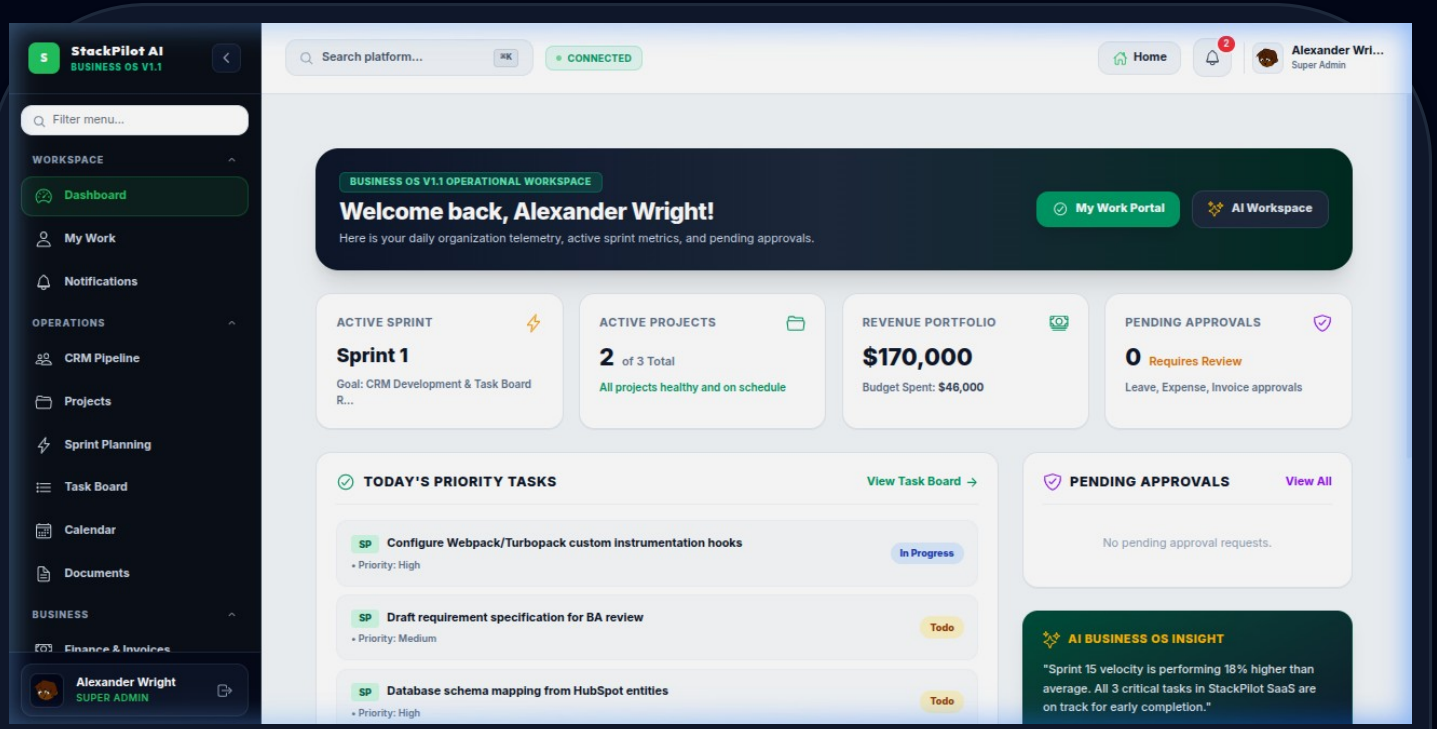


KEY CAPABILITIES

- **UAT Test Case Writer**
  Instantly draft comprehensive test scenarios directly from feature requirements.
- **Bug Reporter & Triage**
  Scaffolds standardized bug reports with reproduction steps and severity tags.
- **Project Scope & Cost Estimator**
  Calculates accurate budget, timeline, and resource estimates from project specs.
- **Client Communication Composer**
  Generates polished client update emails and milestone progress briefs.

EXECUTIVE OVERSIGHT

Unified Operations & Financial Telemetry



KEY CAPABILITIES

- **Revenue Portfolio Telemetry**
  Live metrics tracking total portfolio value, active retainers, and budget spent.
- **Today's Priority Focus**
  Automated task allocation mapped to active sprints and team assignments.
- **AI Velocity Telemetry**
  Real-time burndown calculations and delivery confidence scoring.
- **Cross-Functional Approvals**
  Pending approval queues for milestone sign-offs, releases, and expenses.

INFRASTRUCTURE & ENGINEERING

High-Performance Technical Architecture

300MS DEBOUNCE

Universal Search

Debounced search engine across tasks, clients, documents, and logs without UI stutters.

EXPRESS & IN-MEMORY

Bimodal Persistence

Gracefully falls back to browser local persistence if backend network connection drops.

VITE + REACT 19 + TS

Modern Frontend

Ultra-fast HMR (< 200ms) with full strict TypeScript typing and Tailwind v4.

FRAMER MOTION 12

Fluid Animations

Hardware-accelerated layout transitions delivering a native desktop app feel.

ENTERPRISE TRUST

Security, Privacy & Compliance Architecture

ENTERPRISE SECURITY VERIFICATION CHECKLIST

✔️ **Strict CORS & Origin Allowlist**
Backend configured with strict origin protection against unauthorized API requests.

✔️ **Security Middleware Suite**
Equipped with helmet headers, express-rate-limit, and strict payload sanitization.

✔️ **Token-Based Authentication**
Stateless JWT authentication backed by bcryptjs cryptographic salted password hashing.

✔️ **Immutable Audit Logs**
System logs all critical actions (auth, create, update, delete) with IP and timestamp telemetry.

✔️ **WCAG AA Accessibility**
High-contrast emerald-on-dark slate color palette ensuring compliance for all team members.

PRODUCTION-READY & QA AUDITED

Deploy StackPilot AI in Your Enterprise

Ready for self-hosted container deployment, cloud VPC, or managed SaaS instances. Accelerate your engineering delivery and streamline operations today.

- **Product Version: v1.0.6 Enterprise Edition**
- **Source Repository: saravanaselvi2705/stackpilot-ai**
- **Contact & Sales: creovixstack@gmail.com / sales@stackpilot.ai**
- **Website & Demo: https://stackpilot.ai**